

Lección 17

M. Siddikov

May 13, 2026

Plan de trabajo

►Formulación hamiltoniana

- Transformación de Legendre
- Ecuaciones de Hamilton
- Corchete de Poisson

►Nuestro Syllabus:

FIS-310:

UNIDAD 5. Formulación Hamiltoniana

- 5.1. Transformaciones de Legendre
- 5.2. Ecuaciones de Hamilton
- 5.3. Ecuaciones de Hamilton de la Partícula Relativista
- 5.4. Corchete de Poisson
- 5.5. Transformaciones Canónicas
- 5.6. Ecuaciones de Hamilton-Jacobi
- 5.7. Teoría de perturbaciones

FIS-345:

Unidad IV: Formulación Hamiltoniana General

- Transformaciones canónicas
- Variables de ángulo y acción. Invariantes Adiabáticos

Unidad V: Teoría de perturbaciones y pequeñas oscilaciones

- Teoría canónica de perturbaciones
- Caos en sistemas Hamiltonianos

FIS-426:

4) Formulación Hamiltoniana General

- Paréntesis de Poisson
- Transformaciones canónicas
- Variables de ángulo y acción
- Teoría de perturbaciones
- Caos en sistemas Hamiltonianos

Hamiltonian formulation (summary)

- Use momenta p_a and coordinates q_a to describe system
— "vector" in phase space
- Dynamics is described by Hamiltonian H , related to L via Legendre's transformation

$$H \equiv \sum_a p_a \dot{q}_a - L, \quad L = \sum_a p_a \frac{\partial H}{\partial p_a} - H, \quad p = \frac{\partial L}{\partial \dot{q}_a}$$

Hamilton's (Canonical) equations

$$\dot{p}_a = -\frac{\partial H(p, q, t)}{\partial q_a} = [H, p_a], \quad \dot{q}_a = \frac{\partial H(p, q, t)}{\partial p_a} = [H, q_a], \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.$$

If the particle follows the trajectory $\{p_a(t), q_a(t)\}$, then change of *any* function $f(p, q, t)$ along the trajectory is

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_a \left(\frac{\partial f}{\partial q_a} \dot{q}_a + \frac{\partial f}{\partial p_a} \dot{p}_a \right) = \frac{\partial f}{\partial t} + \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) = \frac{\partial f}{\partial t} + [H, f]$$
$$[H, f] \equiv \sum_a \left(\frac{\partial H}{\partial p_a} \frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a} \frac{\partial f}{\partial p_a} \right) \quad (= \text{Poisson bracket})$$

$\Rightarrow H$ plays the role of generator for translations $t \rightarrow t + t_0$ (recall **Lecture 13**)

Poisson's theorem

– We say that $f(p, q)$ is integral of motion, if $df/dt = 0$

* If $f(p, q)$ does not depend explicitly on time ($\partial f/\partial t = 0$), then $[H, f] \equiv 0$

If $f(p, q)$, $g(p, q)$ are integrals of motion, then their Poisson bracket $h = [f, g]$ is also an integral of motion

- For any system with $N < \infty$ degrees of freedom, the number of integrals of motion is finite. After several iterative application of procedure, the “new” integrals of motion h_i either will start vanishing, or reduce to linear combination of known integrals

Lecture 13:

– For any continuous symmetry of action, there is its own integral of motion

$$Q_a \in \{f, g, \dots\}$$

* These Noether invariants (integrals of motion) play the role of generators of corresponding transformation

$\Rightarrow$ Poisson's theorem = claim that **generators form Lie algebra**, i.e. the set of transformations which don't change action should form a group

Poisson's theorem (proof)

First we'll assume that f, g do not depend on time

– They are integrals of motion, so $[H, f] = [H, g] = 0$

– Need to show that $[H, h] = 0$ for $h = [f, g]$

Jacobi's identity:

$$[[f, g], h] + [[h, f], g] + [[g, h], f] = 0$$

Replace $h \rightarrow H$ (hamiltonian)

$$\left[\underbrace{[f, g]}_h, H \right] + \left[\underbrace{[H, f]}_{=0}, g \right] + \left[\underbrace{[g, H]}_{=0}, f \right] = 0$$

$\Rightarrow [H, h] = 0$, which finishes the proof for this special case

Poisson's theorem (proof)

- Generalize for the case when f, g depend on time
 - They are integrals of motion, so

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [H, f] = 0, \quad \frac{dg}{dt} = \frac{\partial g}{\partial t} + [H, g] = 0$$

- Need to show that $dh/dt = \partial_t h + [H, h] = 0$

$$\begin{aligned} \frac{dh}{dt} &= \frac{d}{dt} [f, g] = \left[\frac{\partial f}{\partial t}, g \right] + \left[f, \frac{\partial g}{\partial t} \right] + [H, [f, g]] = \\ &= \left[\frac{\partial f}{\partial t}, g \right] + \left[f, \frac{\partial g}{\partial t} \right] + [[H, f], g] + [f, [H, g]] = \\ &= \left[\frac{df}{dt}, g \right] + \left[f, \frac{dg}{dt} \right] = 0 \end{aligned}$$

Q.E.D.

Examples

Assume that a particle moves in some potential field. **Demonstrate that:**

- 1) If M_x and M_y are integrals of motion, then M_z also must be integral of motion
 - 2) If M_x and p_y are integrals of motion, then p_z also must be integral of motion
 - 3) If M_x , M_y and p_z are integrals of motion, then all the remaining components M_z, p_x, p_y also must be integrals of motion
- As You may guess, we **may use Poisson's theorem** and identities

$$[M_a, p_b] = -\epsilon_{abj} p_j, \quad [M_a, M_b] = -\epsilon_{abc} M_c$$

(we'll prove them in a couple of slides)

Reminder: Poisson bracket & its properties

- Evaluation of the Poisson bracket using its definition is well-suited for *Mathematica*
 - Not very optimal method for manual evaluation: many terms vanish.
 - Can use simple (yet very powerful) rules which allow to simplify evaluations
-

(below $c_i = \text{const}$, f_i, g are some functions, q_a are coordinates, p_a are momenta)

$$[f, g] = -[g, f], \quad [f, c_i] = 0$$

$$[c_1 f_1 + c_2 f_2, g] = c_1 [f_1, g] + c_2 [f_2, g]$$

$$[f_1 f_2, g] = [f_1, g] f_2 + f_1 [f_2, g]$$

$$[f, q_a] = \frac{\partial f}{\partial p_a}, \quad [f, p_a] = -\frac{\partial f}{\partial q_a}, \quad [q_a, q_b] = [p_a, p_b] = 0, \quad [p_a, q_b] = \delta_{ab},$$

$$[[f, g], h] + [[h, f], g] + [[g, h], f] = 0 \quad (\text{Jacobi identity})$$

If $f(p)$ and $g(q)$ are two functions of momenta p and coordinates q respectively, then

$$[f, g] = \sum_a \frac{\partial f(p)}{\partial p_a} \frac{\partial g(q)}{\partial q_a}$$

Basic examples

$$[\sin p, q] = \frac{\partial \sin p}{\partial p} = \cos p,$$

$$[\sin p, p^2] = 2p [\sin p, p] = -2p \frac{\partial \sin p}{\partial q} = 0,$$

$$[\sin p, q^2] = 2q [\sin p, q] = 2q \frac{\partial \sin p}{\partial p} = 2q \cos p,$$

$$\begin{aligned} [\sin p, p^2 + 2q^2] &= [\sin p, p^2] + 2 [\sin p, q^2] \\ &= 4q \cos p \end{aligned}$$

–2D example: assume q_1, q_2 are coordinates and p_1, p_2 are momenta:

$$[\sin p_1, q_1] = \frac{\partial \sin p_1}{\partial p_1} = \cos p_1,$$

$$[\sin p_1, q_2] = \frac{\partial \sin p_1}{\partial p_2} = 0,$$

More sophisticated examples

Assume that q_a are generalized coordinates, p_a are corresponding momenta, $M_a = \epsilon_{abc} q_b p_c$ are components of the angular momentum, and $\mathbf{M}^2 = \sum_a M_a M_a$

$$[M_a, p_b] = [\epsilon_{aij} q_i p_j, p_b] = \epsilon_{aij} \left(\underbrace{[q_i, p_b]}_{-\delta_{ib}} p_j + q_i \underbrace{[p_j, p_b]}_{=0} \right) = -\epsilon_{abj} p_j$$

$$[M_a, q_b] = [\epsilon_{aij} q_i p_j, q_b] = \epsilon_{aij} \left(\underbrace{[q_i, q_b]}_{=0} p_j + q_i \underbrace{[p_j, q_b]}_{=\delta_{jb}} \right) \epsilon_{ajb} q_j = -\epsilon_{abj} q_j$$

$$\begin{aligned} [M_a, M_b] &= [\epsilon_{aij} q_i p_j, M_b] = \epsilon_{aij} \left(\underbrace{[q_i, M_b]}_{-\epsilon_{ibk} q_k} p_j + q_i \underbrace{[p_j, M_b]}_{-\epsilon_{jbk} p_k} \right) \\ &= -\epsilon_{aij} \epsilon_{ibk} q_k p_j - \underbrace{\epsilon_{aij} \epsilon_{jbk} q_i p_k}_{(ijk) \rightarrow (kij)} = \underbrace{(\epsilon_{aij} \epsilon_{ibk} + \epsilon_{aki} \epsilon_{ibj})}_{\epsilon_{abs} \epsilon_{skj}} q_k p_j = -\epsilon_{abs} M_s \equiv -\epsilon_{abc} M_c \end{aligned}$$

$$[M^2, M_z] = 2 \sum_a M_a [M_a, M_z] = -2 \sum_{aj} \epsilon_{azj} M_a M_j = 0 \quad \text{due to } \epsilon_{azj} = -\epsilon_{jza}$$

Example 3

The particle moves in the central field $U(r)$. Demonstrate that all components M_a of angular momentum are integrals of motions

$$\frac{\partial M_a}{\partial t} = 0 \Rightarrow \text{have to show that } [H, M_a] = 0$$

$$H = \frac{p^2}{2m} + U(r), \quad r = \sqrt{x^2 + y^2 + z^2}$$

$$[H, M_z] = \frac{1}{2m} [p^2, M_z] + [U(r), M_z],$$

$$[p^2, M_z] = 2 \sum_a p_a [p_a, M_z] = -2 \sum_{aj} \epsilon_{azj} p_a p_j = 0$$

due to antisymmetry of $\epsilon_{azj} = -\epsilon_{jza}$

$$\begin{aligned} [U(r), M_z] &= [U(r), q_x p_y - q_y p_x] = \\ &= \underbrace{[U(r), q_x]}_{=0} p_y + \underbrace{q_x}_{=x} \underbrace{[U(r), p_y]}_{-\partial U / \partial y} - \underbrace{q_y}_y \underbrace{[U(r), p_x]}_{-\partial U / \partial x} - p_x \underbrace{[q_y, U(r)]}_{=0} = \\ &= -U'(r) \frac{x y}{r} + U'(r) \frac{x y}{r} = 0 \end{aligned} \quad \Rightarrow [H, M_z] = 0.$$

Canonical equations - symplectic notations

- In the modern literature many authors discuss canonical formalism using the so-called *symplectic* notations. These notations allow to simplify some considerations, especially related to canonical transformations etc.

Our goal: Learn/understand this symplectic language.

I will follow 3 books:

- Goldstein from bibliography
- V. Arnold from bibliography
- B. Shutz, "*Geometrical methods of mathematical physics*", Sections 5.4-5.10

Canonical equations - symplectic notations

Hamilton's (Canonical) equations

$$\dot{p}_a = -\frac{\partial H(p, q, t)}{\partial q_a}, \quad \dot{q}_a = \frac{\partial H(p, q, t)}{\partial p_a}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \quad (1)$$

symmetric form : $\dot{q}_a = [H, q_a]$, $\dot{p}_a = [H, p_a]$

- Let's introduce a "joined" notation $\boldsymbol{\eta}$ for vector in $2N$ **phase space** with components

$$\boldsymbol{\eta} = \begin{pmatrix} q_1 \\ \vdots \\ q_N \\ p_1 \\ \vdots \\ p_N \end{pmatrix}, \quad \Rightarrow \quad \text{equation of motion : } \dot{\boldsymbol{\eta}} = [H, \boldsymbol{\eta}],$$

We completely erase any difference between coordinates and momenta, treat them on equal footing as components of vector $\boldsymbol{\eta}$

*Looka "dangerous": potentially can mix $\mathbf{q}$ and $\mathbf{p}$. But may be useful in some problems

**First we'll see some problems to understand better the concept of the phase space

Phase space

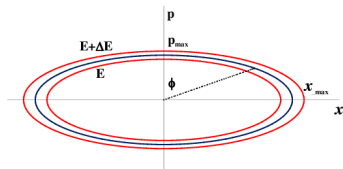
Hypothetical $2N$ -dimensional space, in which “coordinates” are generalized coordinates and generalized momenta of the system

Example for 1D harmonic oscillator:

$$x(t) = A \cos(\omega t + \phi), \quad p(t) = -m A \omega \sin(\omega t + \phi), \quad \omega = \sqrt{k/m}.$$

$$\eta = \begin{pmatrix} x(t) \\ p(t) \end{pmatrix} = A \begin{pmatrix} \cos(\omega t + \phi) \\ -m \omega \sin(\omega t + \phi) \end{pmatrix}$$

$$\frac{p^2}{2mE} + \frac{x^2}{2E/k} = 1$$



Example - Harmonic oscillator in 3D

The hamiltonian of 3D harmonic oscillator is given by

$$H = \frac{\mathbf{p}^2}{2m} + \frac{k\mathbf{q}^2}{2} = \frac{1}{2m} \left(\sum_{a=1}^3 \mathbf{p}_a^2 \right) + \frac{k}{2} \left(\sum_{a=1}^3 \mathbf{q}_a^2 \right)$$

What is the symmetry group of this hamiltonian?

(*"What symmetries has its hamiltonian?"*)

Example - Harmonic oscillator in 3D

$$H = \frac{\mathbf{p}^2}{2m} + \frac{k\mathbf{q}^2}{2} = \frac{1}{2m} \left(\sum_{a=1}^3 \eta_a^2 \right) + \frac{k}{2} \left(\sum_{a=1}^3 \eta_a^2 \right)$$

–(Naive) Symmetries based on Lagrangian picture:

$$\mathbf{q}_a \rightarrow R_{ab} \mathbf{q}_b, \quad \mathbf{p}_a \rightarrow R_{ab} \mathbf{p}_b, \quad \mathbf{R}^T \mathbf{R} = 1 \quad (\text{orthogonal matrix, } \in \text{SO}(3))$$

–Let's change variables:

$$\mathbf{Q} = \mathbf{p} + i\sqrt{km} \mathbf{q}, \quad \mathbf{Q}^\dagger = \mathbf{p} - i\sqrt{km} \mathbf{q}$$

$$H = \frac{1}{2m} \left(\sum_{a=1}^3 \mathbf{Q}_a^\dagger \mathbf{Q}_a \right)$$

Can see that symmetry group is MUCH larger,

$$\mathbf{Q}_a \rightarrow U_{ab} \mathbf{Q}_b, \quad \mathbf{Q}_a^\dagger \rightarrow \left(\mathbf{Q}^\dagger U^\dagger \right)_a = \mathbf{Q}_b^\dagger U_{ba}^\dagger, \quad U^\dagger U = 1 \quad (\text{unitary matrix, } \in \text{SU}(3))$$

–In special case when U is real-valued, it does not mix real and imaginary parts of $\mathbf{Q}$, so this reduces to $\text{SO}(3)$

–In general U is complex-valued, so it mixes real and imaginary parts of $\mathbf{Q}$ (momenta $\mathbf{p}$ and coordinates $\mathbf{q}$)

Example - Harmonic oscillator in 3D

$$H = \frac{1}{2m} \left(\sum_{a=1}^3 Q_a^\dagger Q_a \right), \quad \mathbf{Q} = \mathbf{p} + i\sqrt{km} \mathbf{q}, \quad \mathbf{Q}^\dagger = \mathbf{p} - i\sqrt{km} \mathbf{q}$$

Symmetry group:

$$Q_a \rightarrow U_{ab} Q_b, \quad Q_a^\dagger \rightarrow \left(Q^\dagger U^\dagger \right)_a = Q_b^\dagger U_{ba}^\dagger, \quad U^\dagger U = 1 \quad (\text{unitary matrix, } \in \text{SU}(3))$$

- For symmetries which only mix coordinates $\mathbf{q}$ (and time) we may apply Noether theorem, deduce integrals of motion (Noether invariants) ...
- Now we have symmetry which mixes coordinates and momenta $\mathbf{p}$, so we need to understand this better
- We'll try to analyze such transformations in general
Goal : Find the most general transformation which can mix $\mathbf{p}$ and $\mathbf{q}$, but keeps canonical equations
- * For a moment we'll return back to conventional (non-symplectic) notations, and will explicitly distinguish coordinates $\mathbf{q}$ and momenta $\mathbf{p}$

Control problem

$$\eta = \begin{pmatrix} q_1 \\ \dots \\ q_N \\ p_1 \\ \dots \\ p_N \end{pmatrix}, \quad \Rightarrow \quad \text{equation of motion : } \dot{\eta} = [H, \eta],$$

Using conventional Poisson brackets

$$[q_a, q_b] = 0, \quad [p_a, p_b] = 0, \quad [p_a, q_b] = -[q_a, p_b] = \delta_{ab},$$

find explicitly the value of Poisson bracket (in symplectic notations)

$$[\eta_a, \eta_b] = J_{ab}$$

Help:

- What is the dimension of matrix J_{ab} ?
- What can we say about its symmetries?
- What elements are not zero?

Control problem (result)

$$[\eta_a, \eta_b] = J_{ab}, \quad \Leftrightarrow \quad [\boldsymbol{\eta}, \boldsymbol{\eta}] = \boldsymbol{J}$$

$$\boldsymbol{J} = \begin{pmatrix} & & & 1 & 0 & 0 \\ & 0 & & 0 & \dots & 0 \\ -I_N & & & 0 & 0 & 1 \\ & -1 & 0 & 0 & & \\ & 0 & \dots & 0 & & 0 \\ & 0 & 0 & -1 & & \end{pmatrix}$$

The matrix J_{ab} can also be used to rewrite the Poisson brackets

$$[f, g] = \left(\frac{\partial f}{\partial \boldsymbol{\eta}} \right)^T \boldsymbol{J} \frac{\partial g}{\partial \boldsymbol{\eta}} \equiv \sum_{ab} \frac{\partial f}{\partial \eta_a} J_{ab} \frac{\partial g}{\partial \eta_b}$$

$$\dot{\eta}_a = [H, \eta_a] = J_{ab} \frac{\partial H}{\partial \eta_b}, \quad \Leftrightarrow \quad \dot{\boldsymbol{\eta}} = [H, \boldsymbol{\eta}] = \boldsymbol{J} \frac{\partial H}{\partial \boldsymbol{\eta}}$$

–The peculiar antisymmetric structure of J_{ab} allows us to formulate the mechanics in terms of the so-called **symplectic forms**

Canonical equations - symplectic form

Now we'll try to rewrite everything in terms of the canonical **symplectic forms** which are used in the literature.

– Formal definition: differential form of rank n is a linear homogeneous function of n arguments, antisymmetric with respect to permutation of any pair of its arguments

$$\omega(\mathbf{V}_1, \dots, \mathbf{V}_i, \dots, \mathbf{V}_j, \dots, \mathbf{V}_n) = -\omega(\mathbf{V}_1, \dots, \mathbf{V}_j, \dots, \mathbf{V}_i, \dots, \mathbf{V}_n)$$

– If You are not familiar with 1-forms, 2-forms, ..., just consider them as tensors with all covariant indices (opposite to vectors) and antisymmetrized over all indices:

* 1-forms: $dx^a = \{dx, dy, dz\}$; $A = \sum_a A_a dx^a$

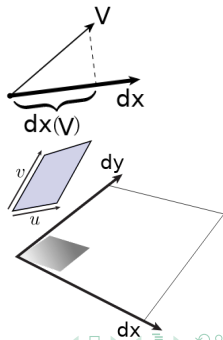
* 2-forms: $A \wedge B \equiv A \otimes B - B \otimes A = \sum_{a < b} (A_a B_b - B_a A_b) dx^a \wedge dx^b$

$$dx^a \wedge dx^b = \{dx \wedge dy, dy \wedge dz, dz \wedge dx\}$$

Sometimes forms are considered as “projectors”:

1-form dx convoluted with any vector $\mathbf{V}$ gives its component V^x (projection on direction x)

2-form $dx \wedge dy$ convoluted with vectors $\mathbf{u}, \mathbf{v}$ gives area of parallelepiped constructed from projections of $\mathbf{u}, \mathbf{v}$ on Oxy plane, etc



Cartan's notations

As You may have noticed, instead of $\omega_x, \omega_y, \dots$ we use notation $dx, dy, \dots$ which “look like” differentials

–Not a coincidence: if we make arbitrary transformation of coordinates $x \rightarrow \xi$, then a form $A_a dx^a$ transforms as

$$A_a dx^a = \left(A_a \frac{\partial x^a}{\partial \xi^b} \right) d\xi^b = \tilde{A}_b d\xi^b$$

Similarly, for vectors instead of base vectors $\mathbf{e}_x, \mathbf{e}_y, \dots$ we may use notation $\partial_x, \partial_y, \dots$

–if we make arbitrary transformation of coordinates $x \rightarrow \xi$, then a vector $V^a \partial_{x_a}$ transforms as

$$V^a \frac{\partial}{\partial x^a} = \left(V^a \frac{\partial \xi^b}{\partial x^a} \right) \frac{\partial}{\partial \xi^b} = \tilde{V}^b \frac{\partial}{\partial \xi^b}$$

Note that now we speak about any transformation of coordinates. We do not require that it is orthogonal, does not mix “ q ” and “ p ”, etc.

Cartan's notations and derivatives

Assume that $x^a(\lambda)$ is some curve, and $V^a = dx^a/d\lambda$ is tangent vector.

–For any function $f(x)$, its change (derivative) along the curve satisfies

$$\frac{df}{d\lambda} = \frac{dx^a}{d\lambda} \frac{\partial f}{\partial x^a} = V^a \frac{\partial f}{\partial x^a} \equiv \hat{\mathbf{V}}f \equiv \mathcal{L}_V f$$

We call $\mathcal{L}_V \equiv \hat{\mathbf{V}} = V^a \frac{\partial}{\partial x^a}$ a **Lie derivative**: operator in which $\frac{\partial}{\partial x^a}$ are considered as partial derivatives (not as shortcut to memorize transformation rules)

–Similarly can define **exterior derivative** d which acts on scalars, forms:

$$df = \frac{\partial f}{\partial x^a} dx^a,$$

$$d(A_a dx^a) = \sum_{a,b} \left(\frac{\partial A_a}{\partial x^b} dx^b \right) \wedge dx^a = \sum_{a < b} \left(\frac{\partial A_a}{\partial x^b} - \frac{\partial A_b}{\partial x^a} \right) dx^b \wedge dx^a$$

Lemma

Simplectic form

The 2-form defined as

$$\omega = \sum dq^a \wedge dp^a \equiv - \sum dp^a \wedge dq^a \equiv \sum J_{ab} d\eta^a \otimes d\eta^b = \frac{1}{2} \sum J_{ab} d\eta^a \wedge d\eta^b$$
$$J \equiv \begin{pmatrix} & I_N \\ -I_N & \end{pmatrix}$$

–The form ω is closed: $d\omega = ddq^a \wedge dp^a - dq^a \wedge dp^a = 0$.

Poincaré theorem:

- If the phase space is single-connected, then ω is *exact* (exists 1-form h such that $\omega = dh$)
- Assume $(q_a, p_a) = \{q_a(t), p_a(t)\}$ is the trajectory. We may define the tangent vector to trajectory

$$U = \dot{q}_a(t) \frac{\partial}{\partial q^a} + \dot{p}_a \frac{\partial}{\partial p^a}$$

Then for the total time derivative of *any* scalar function $f(t, q, p)$ along the trajectory we may write

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + Uf = \frac{\partial f}{\partial t} + \underbrace{\mathcal{L}_U f}$$

Lie derivative

Lemma

The form ω satisfies

$$\mathcal{L}_U \omega = 0,$$

Proof:

► $\forall U, \omega$ we have Cartan's formula (also known as Cartan's theorem):

$$\mathcal{L}_U \omega = [\mathbf{d}\omega(U) + \mathbf{d}(\omega \cdot U)];$$

since $\mathbf{d}\omega = 0$, then

$$\begin{aligned}\mathcal{L}_U \omega &= \mathbf{d}(\omega \cdot U) = \mathbf{d}(\dot{q}_a(t) dp^a - \dot{p}_a dq^a) \\ &= \mathbf{d} \left(\frac{\partial H}{\partial p^a} dp^a + \frac{\partial H}{\partial q^a} dq^a \right) = \mathbf{d}dH \equiv 0\end{aligned}$$

Q.E.D.

- Also conclude that $\omega \cdot U \equiv \mathbf{d}H$. We'll call any field U which has this property *Hamiltonian vector field*.
- Since the symplectic form $\omega = dq \wedge dp$ does not depend explicitly on time, $\partial_t \omega = 0$, this lemma implies that $d\omega/dt = 0$ along the particle's trajectory.

Lemma

For any hamiltonian vector field

$$U = \dot{q}_a(t) \frac{\partial}{\partial q^a} + \dot{p}_a \frac{\partial}{\partial q^a}$$

it is always possible to find a function $H(p, q)$, such that

$$\omega(U) = dH \quad (1)$$

(non-symplectic language: for any trajectory $q(t), p(t)$ it is possible to construct Hamiltonian, such that U satisfies equations of motion)

Proof:

We've shown that $d\omega(U) = 0$, i.e. $\omega(U)$ is closed 1-form. If we assume that phase space is single-connected, applying Poincaré's theorem, can see that it must be exact, i.e. $\exists H$, such that (1) is valid.

Lemma

All hamiltonian vector fields form Lie algebra, i.e. if $\mathbf{U}, \mathbf{V}$ are hamiltonian vector fields, then Lie bracket

$$\mathbf{W} = [\mathbf{U}, \mathbf{V}] = \left(U^i \frac{\partial V^j}{\partial x^i} - V^i \frac{\partial U^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}$$

is also hamiltonian vector field.

Proof:

By definition

$$\mathcal{L}_U \omega = \mathcal{L}_V \omega = 0 \quad (1)$$

$$\mathcal{L}_W \omega = \mathcal{L}_{[U, V]} \omega = (\mathcal{L}_U \mathcal{L}_V - \mathcal{L}_V \mathcal{L}_U) \omega = 0, \quad \text{Q.E.D.}$$

(How should we reformulate this property in non-symplectic language ?)

Canonical equations - symplectic form

► We may use symplectic form to relate vectors and 1-forms:

– If $\bar{U}$ is (any) vector field, then $\tilde{U} = \omega(\bar{U})$ is 1-form

– If $\tilde{U}$ is (any) 1-form, we define correspondent vector field $\bar{U}$ as solution of $\tilde{U} = \omega(\bar{U})$.

– Relation of $\tilde{U}$ and $\bar{U}$:

$$\tilde{U} = f_a dq^a + g_a dp^a$$

$$\bar{U} = g_a \frac{\partial}{\partial q^a} - f^a \frac{\partial}{\partial p^a}$$

Canonical equations - symplectic form

Assume that f, g are 2 arbitrary scalar functions, and $X_f \equiv \overline{df}$, $X_g \equiv \overline{dg}$ are the vector fields obtained applying ω to 1-forms df, dg . Then the Poisson bracket

$$[f, g] \equiv \omega(X_f, X_g) = df(X_g) = dg(X_f)$$

Proof:

$$df = \frac{\partial f}{\partial p^a} dp^a + \frac{\partial f}{\partial q^a} dq^a$$

$$dg = \frac{\partial g}{\partial p^a} dp^a + \frac{\partial g}{\partial q^a} dq^a$$

$$X_f = \frac{\partial f}{\partial p^a} \frac{\partial}{\partial q^a} - \frac{\partial f}{\partial q^a} \frac{\partial}{\partial p^a}$$

$$X_g = \frac{\partial g}{\partial p^a} \frac{\partial}{\partial q^a} - \frac{\partial g}{\partial q^a} \frac{\partial}{\partial p^a}$$

$$\omega(X_f, X_g) = \frac{\partial f}{\partial p^a} \frac{\partial g}{\partial q^a} - \frac{\partial f}{\partial q^a} \frac{\partial g}{\partial p^a} = [f, g]$$

Q.E.D.

Canonical equations - symplectic form

► Evolution for any quantity Q :

$$\frac{dQ}{dt} = \frac{\partial Q}{\partial t} + [H, Q]$$

may be rewritten as

$$\frac{dQ}{dt} = \frac{\partial Q}{\partial t} + X_H Q = \frac{\partial Q}{\partial t} + \mathcal{L}_{X_H} Q$$

where $\overline{X_H} = \omega(\mathbf{d}H)$

– If Q does not depend explicitly on t , then it is integral of motion iff

$$\mathcal{L}_{X_H} Q = 0$$

Canonical equations - symplectic form

► Evolution for any quantity Q :

$$\frac{dQ}{dt} = \frac{\partial Q}{\partial t} + [H, Q]$$

might be rewritten as

$$\frac{dQ}{dt} = \frac{\partial Q}{\partial t} + X_H Q = \frac{\partial Q}{\partial t} + \mathcal{L}_{X_H} Q$$

where $\overline{X_H} = \omega(\mathbf{d}H)$

– If Q does not depend explicitly on t , then it is integral of motion iff

$$\mathcal{L}_{X_H} Q = 0$$

Canonical equations - symplectic form

- We can see that in symplectic formalism we can work with completely coordinate-free notations.
 - We don't even need to distinguish coordinates and momenta, tacitly assuming that they are just components of vector η
- * This can look “dangerous”, since potentially can mix them (q and p)
- * For a moment we'll return back to conventional (non-symplectic) notations, and will explicitly distinguish coordinates q and momenta p

Canonical transformations (motivation)

Earlier discussion:

- Sometimes non-Cartesian coordinates simplify drastically analysis of the problem
 - Example: Kepler problem → spherical coordinates
- Important fact which justifies use of curvilinear coords:
 - Covariance of Euler-Lagrange equations with respect to substitutions of variables $q_a \rightarrow q_a(r_1, r_2, r_3)$

$$\frac{\partial L}{\partial q_a} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_a} \right) = 0 \quad \Leftrightarrow \quad \frac{\partial L}{\partial r_a} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}_a} \right) = 0$$

► Hamiltonian approach:

- Coordinates and momenta contribute on equal footing, as “coordinates” $\eta = \{q_a, p_a\}$ in phase space
 - * Transformations Cartesian ↔ spherical ↔ ... mix only coordinates q_a (first N components of η) with each other
 - * Potentially could also consider transformations which also mix coordinates q_a and momenta p_a ???

Canonical transformations (introduction)

Caution:

- We still want to have the same form of canonical equations

$$\dot{P}_a = -\frac{\partial \tilde{H}(P, Q, t)}{\partial Q_a}, \quad \dot{Q}_a = \frac{\partial \tilde{H}(P, Q, t)}{\partial P_a}, \quad (1)$$

or in symplectic form

$$\dot{\eta} = [H, \eta] = J \frac{\partial H}{\partial \eta}, \quad J = \begin{pmatrix} & I_N \\ -I_N & \end{pmatrix} \quad (2)$$

in order to be able to interpret new variables P , Q as coordinates and momenta

- If we consider *arbitrary* transformation

$$Q_a = Q_a(q_1, \dots, q_N, p_1, \dots, p_N, t), \\ P_a = P_a(q_1, \dots, q_N, p_1, \dots, p_N, t).$$

then clearly (1,2) might be not valid (get extra factors with derivatives).

Canonical transformations

Canonical transformation

Transformation $(q, p, t) \rightarrow (Q, P, t)$ is canonical if it preserves the form of canonical (Hamilton) equations, *i.e.*

$$\dot{P}_a = -\frac{\partial \tilde{H}(P, Q, t)}{\partial Q_a}, \quad \dot{Q}_a = \frac{\partial \tilde{H}(P, Q, t)}{\partial P_a}, \quad (1)$$

► Some trivial properties:

- The identity transformation $\{P_a = p_a, Q_a = q_a\}$ is canonical
- If the transformation is canonical, then its inverse is also canonical
- Superposition of two canonical transformations is also canonical

Lemma

Earlier we discussed special (limited) case of transformations which affects only coordinates,

$$q_\alpha \rightarrow q_\alpha(Q_1, \dots, Q_N), \quad Q_\alpha \rightarrow Q_\alpha(q_1, \dots, q_N)$$

don't mix them with momenta, e.g. Cartesian $\leftrightarrow$ Spherical $\leftrightarrow$ Cylindrical. We'll call them **point** transformations

Lemma

All point transformations of coordinates are canonical

In simple words:

We know that lagrangian formalism and Euler-Lagrange equations are valid in any curvilinear coordinates (spherical, polar, cartesian). For this reason we expect that Hamiltonian approach should give equivalent results in any of these frames. Now we need to show that this is indeed so.

Proof of lemma

– Assume (q_a, p_a) are variables before transformation, and (Q_a, P_a) are variables after transformation

– Canonical equations:

$$\dot{q}_a = \frac{\partial H}{\partial p_a}, \quad \dot{p}_a = -\frac{\partial H}{\partial q_a}$$

– Apply chain rule to the first equation, recalling that $q_\alpha(Q_1, \dots, Q_N)$

$$\dot{q}_a = \sum_b \dot{Q}_b \frac{\partial q_a}{\partial Q_b} = \frac{\partial H}{\partial p_a} \quad (1)$$

– Now recall that $Q_\alpha(q_1, \dots, q_N)$ and work with chain rule again:

$$\delta_b^c = \frac{\partial Q_c}{\partial Q_b} = \sum_b \frac{\partial q_a}{\partial Q_b} \frac{\partial Q_c}{\partial q_a} = \delta_b^c \quad (2)$$

$$\sum_{ab} \dot{Q}_b \frac{\partial q_a}{\partial Q_b} \frac{\partial Q_c}{\partial q_a} = \sum_b \dot{Q}_b \delta_b^c = \dot{Q}_c \quad (3)$$

Now let's multiply both parts of (1) on matrix $\frac{\partial Q_c}{\partial q_a}$ and make summation over a (the idea is to try to extract $\dot{Q}_a$ using (2,3))

Proof of lemma (II)

$$\Rightarrow \dot{Q}_c = \sum_a \frac{\partial H}{\partial p_a} \left(\frac{\partial Q_c}{\partial q_a} \right) := \frac{\partial H}{\partial P_c} \quad (1)$$

– Expression in right-hand side is what we *expect* for point transformation. If we *assume* that momenta P_a are related to momenta p_a as

$$\frac{\partial P_a}{\partial p_b} = \left(\frac{\partial q_b}{\partial Q_a} \right), \quad \frac{\partial p_a}{\partial P_b} = \frac{\partial Q_b}{\partial q_a} \quad (2)$$

then (1) turns into chain rule, becomes valid identically

Proof of lemma (III)

$$\frac{\partial P_a}{\partial p_b} = \left(\frac{\partial q_b}{\partial Q_a} \right), \quad \frac{\partial p_a}{\partial P_b} = \frac{\partial Q_b}{\partial q_a} \quad (2)$$

–May repeat the same exercise for the second canonical equation:

$$\dot{p}_a = -\frac{\partial H}{\partial q_a} = \sum_b -\frac{\partial H}{\partial Q_b} \frac{\partial Q_b}{\partial q_a} \quad (3)$$

Let's rewrite (2) as

$$\dot{P}_a = \left(\frac{\partial q_b}{\partial Q_a} \right) \dot{p}_b, \quad \Rightarrow \dot{p}_b = \left(\frac{\partial Q_b}{\partial q_a} \right) \dot{P}_b \quad (2')$$

substitute (2') into (3) and multiply both parts onto matrix $\partial q_b / \partial Q_a$:

$$\Rightarrow \dot{P}_b = -\frac{\partial H}{\partial Q_b}$$

as expected for canonical transformation

$\Rightarrow$ If transformation of momenta $P(p)$ agrees with (2), then we can guarantee that point transformation is canonical, Q.E.D.

Canonical transformations of coordinates

Assume that we have a general transformation

$$q = q(Q, P), \quad p = p(Q, P).$$

What conditions it should satisfy in order to be canonical?

Please suggest algorithm which could be used to find such relations.

Canonical transformations of coordinates

$$q = q(Q, P), \quad p = p(Q, P).$$

By definition, canonical transformation should NOT change the form of canonical equations



Our **algorithm of derivation**:

1) Write the canonical equations in coordinates (q, p)

$$\dot{q}_a = -\frac{\partial H}{\partial p_a}, \quad \dot{p}_a = \frac{\partial H}{\partial q_a},$$

2) Rewrite derivatives in terms of variable (Q, P) using chain rules for derivatives

3) Request that all “extra” factors cancel, so that get normal canonical equations in terms of variables (Q, P)

$$\dot{P}_a = -\frac{\partial H(P, Q, t)}{\partial Q_a}, \quad \dot{Q}_a = \frac{\partial H(P, Q, t)}{\partial P_a},$$

– Subtle point: we assume that hamiltonian does not change, *i.e.* just need to substitute variables in “old” hamiltonian (*this is correct for time-independent transformations*):

$$H_{PQ}(P, Q) = H_{pq}(q = q(Q, P), p = p(Q, P))$$

Canonical transformations of coordinates

$$\dot{p}_a = -\frac{\partial H}{\partial q_a}, \quad \dot{q}_a = \frac{\partial H}{\partial p_a}, \quad \dot{P}_a = -\frac{\partial H}{\partial Q_a}, \quad \dot{Q}_a = \frac{\partial H}{\partial P_a}$$

$$\frac{\partial H}{\partial P_a} = \frac{\partial H}{\partial p_b} \frac{\partial p_b}{\partial P_a} + \frac{\partial H}{\partial q_b} \frac{\partial q_b}{\partial P_a}$$

$$\dot{Q}_a = \frac{\partial H}{\partial P_a} = \underbrace{\frac{\partial H}{\partial p_b} \frac{\partial p_b}{\partial P_a}}_{\dot{q}_b} + \underbrace{\frac{\partial H}{\partial q_b} \frac{\partial q_b}{\partial P_a}}_{-\dot{p}_b}$$

$$\dot{Q}_a = \frac{\partial Q_a}{\partial p_b} \dot{p}_b + \frac{\partial Q_a}{\partial q_b} \dot{q}_b$$

$$\Rightarrow \left(\frac{\partial q_b}{\partial P_a} \right)_{Q,P} = - \left(\frac{\partial Q_a}{\partial p_b} \right)_{q,p}, \quad \left(\frac{\partial Q_a}{\partial q_b} \right)_{p,q} = \left(\frac{\partial p_b}{\partial P_a} \right)_{Q,P}$$

Similar analysis of the equation for $\dot{P}_a$ yields

$$\left(\frac{\partial P_b}{\partial p_a} \right)_{q,p} = \left(\frac{\partial q_a}{\partial Q_b} \right)_{Q,P}, \quad \left(\frac{\partial P_a}{\partial q_b} \right)_{p,q} = - \left(\frac{\partial p_b}{\partial Q_a} \right)_{Q,P}$$

Lemma

Poisson bracket $[f, g]$

$$[f, g]_{p,q} = \sum_a \left(\frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q_a} - \frac{\partial f}{\partial q_a} \frac{\partial g}{\partial p_a} \right)$$

$$[f, g]_{P,Q} = \sum_a \left(\frac{\partial f}{\partial P_a} \frac{\partial g}{\partial Q_a} - \frac{\partial f}{\partial Q_a} \frac{\partial g}{\partial P_a} \right)$$

If $(p, q) \rightarrow (P, Q)$ is canonical transformation, then Poisson bracket is invariant, i.e.

$$[f, g]_{p,q} = [f, g]_{P,Q}$$

Canonical transformations of coordinates

Demonstrate that if $(p, q) \rightarrow (P, Q)$ is canonical transformation, then Poisson bracket is invariant, i.e.

$$[f, g]_{p,q} = [f, g]_{P,Q}$$

$$\begin{aligned} [f, g]_{p,q} &= \sum_a \left(\frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q_a} - \frac{\partial f}{\partial q_a} \frac{\partial g}{\partial p_a} \right) \\ \frac{\partial f}{\partial p_a} &= \sum_b \left(\frac{\partial f}{\partial Q_b} \frac{\partial Q_b}{\partial p_a} + \frac{\partial f}{\partial P_b} \frac{\partial P_b}{\partial p_a} \right) \\ \frac{\partial f}{\partial q_a} &= \sum_b \left(\frac{\partial f}{\partial Q_b} \frac{\partial Q_b}{\partial q_a} + \frac{\partial f}{\partial P_b} \frac{\partial P_b}{\partial q_a} \right) \end{aligned}$$

$$\begin{aligned} [f, g]_{p,q} &= \\ \sum_{abc} \left(\frac{\partial f}{\partial P_b} \frac{\partial g}{\partial Q_c} - \frac{\partial g}{\partial P_b} \frac{\partial f}{\partial Q_c} \right) \frac{\partial P_b}{\partial p_a} \frac{\partial Q_c}{\partial q_a} \end{aligned}$$

but we have seen that

$$\left(\frac{\partial Q_a}{\partial q_b} \right)_{p,q} = \left(\frac{\partial p_b}{\partial P_a} \right)_{Q,P}$$

$$\left(\frac{\partial P_a}{\partial q_b} \right)_{p,q} = - \left(\frac{\partial p_b}{\partial Q_a} \right)_{Q,P}$$

$$\begin{aligned} \Rightarrow \frac{\partial P_b}{\partial p_a} \frac{\partial Q_c}{\partial q_a} &= \frac{\partial P_b}{\partial p_a} \frac{\partial p_c}{\partial P_c} = \delta_{bc} \\ \Rightarrow [f, g]_{p,q} &= [f, g]_{P,Q} \end{aligned}$$

► We conclude that Poisson bracket does not change after canonical transformation

Some consequences

$$[f, g]_{p,q} = [f, g]_{P,Q}$$

Why this result is important?

► We know that

$$[q_a, q_b]_{p,q} = 0, \quad [p_a, p_b]_{p,q} = 0, \quad [p_a, q_b]_{p,q} = \delta_{ab},$$

and

$$[Q_a, Q_b]_{P,Q} = 0, \quad [P_a, P_b]_{P,Q} = 0, \quad [P_a, Q_b]_{P,Q} = \delta_{ab},$$

► Assume we know $P_a = P_a(p, q)$ and $Q_b = Q_b(p, q)$. Then for canonical transform

$$[Q_a, Q_b]_{p,q} = 0, \quad [P_a, P_b]_{p,q} = 0, \quad [P_a, Q_b]_{p,q} = \delta_{ab},$$

-usually very easy to check

Example

Demonstrate that the transformation

$$Q = \log \left(\frac{\sin p}{q} \right), \quad P = q \cot p$$

is canonical

► $[Q, Q]_{p,q} = 0$ and $[P, P]_{p,q} = 0$ due to antisymmetry of Poisson's bracket

►

$$\begin{aligned} [P, Q]_{p,q} &= \frac{\partial P}{\partial p} \frac{\partial Q}{\partial q} - \frac{\partial P}{\partial q} \frac{\partial Q}{\partial p} = -\cancel{\frac{q}{\sin^2 p}} \frac{-1}{\cancel{q}} - \cot p \cot p = \\ &= \left(\frac{1}{\sin^2 p} - \cot^2 p \right) = 1. \end{aligned}$$

This finishes the proof

Example 2

Using properties of Poisson's brackets, demonstrate that Galileo's transformation is canonical

$$q_a = Q_a + v_a t, \quad p_a = P_a + m v_a$$

► Just evaluate brackets assuming $v_a = \text{const}$

$$\begin{aligned} [q_a, q_b]_{P,Q} &= [Q_a + v_a t, Q_b + v_b t]_{P,Q} \\ &= [Q_a, Q_b]_{P,Q} + t^2 [v_a, v_b]_{P,Q} + t [Q_a, v_b]_{P,Q} + t [v_a, Q_b]_{P,Q} = 0 \end{aligned}$$

$$\begin{aligned} [p_a, p_b]_{P,Q} &= [P_a + m v_a, P_b + m v_b]_{P,Q} \\ &= [P_a, P_b]_{P,Q} + m^2 [v_a, v_b]_{P,Q} + m [P_a, v_b]_{P,Q} + m [v_a, P_b]_{P,Q} = 0 \end{aligned}$$

$$\begin{aligned} [p_a, q_b]_{P,Q} &= [P_a + m v_a, Q_b + v_b t]_{P,Q} \\ &= \underbrace{[P_a, Q_b]_{P,Q}}_{\delta_{ab}} + m t [v_a, v_b]_{P,Q} + t [P_a, v_b]_{P,Q} + m [v_a, Q_b]_{P,Q} = \delta_{ab} \end{aligned}$$

Thus we have shown that Poisson's brackets satisfy the required condition, which proves that transformation is canonical

Example 3

Demonstrate that for canonical transformation the symplectic form ω is invariant, i.e.

$$\omega = \sum dq^a \wedge dp^a = \sum dQ^b \wedge dP^b$$

Canonical transformations of coordinates

Start with definition:

$$\omega = \sum dq^a \wedge dp^a$$

$$dq^a = \frac{\partial q^a}{\partial Q^c} dQ^c + \frac{\partial q^a}{\partial P^c} dP^c = \frac{\partial P^c}{\partial p^a} dQ^c - \frac{\partial Q^c}{\partial p^a} dP^c$$

$$dp^a = \frac{\partial p^a}{\partial Q^e} dQ^e + \frac{\partial p^a}{\partial P^e} dP^e = -\frac{\partial P^e}{\partial q^a} dQ^e + \frac{\partial Q^e}{\partial q^a} dP^e$$

When making substitution, should take into account that

$$dQ^c \wedge dQ^e = -dQ^e \wedge dQ^c, \quad dP^c \wedge dP^e = -dP^e \wedge dP^c,$$

The constants in front of $dQ \wedge dQ$ and $dP \wedge dP$, given by

$$\frac{\partial P^c}{\partial p^a} \frac{\partial P^e}{\partial q^a}, \quad \frac{\partial Q^c}{\partial p^a} \frac{\partial Q^e}{\partial q^a}$$

vanish after antisymmetrization w.r.t. $c \leftrightarrow e$:

$$\frac{\partial P^c}{\partial p^a} \frac{\partial P^e}{\partial q^a} - \frac{\partial P^e}{\partial p^a} \frac{\partial P^c}{\partial q^a} = [P^c, P^e] = 0, \quad \frac{\partial Q^c}{\partial p^a} \frac{\partial Q^e}{\partial q^a} - \frac{\partial Q^e}{\partial p^a} \frac{\partial Q^c}{\partial q^a} = [Q^c, Q^e] = 0$$

so the terms $dQ \wedge dQ$ and $dP \wedge dP$ won't appear. For the constant in front of $dP^c \wedge dQ^e$ we get

$$\frac{\partial q^a}{\partial Q^e} \frac{\partial Q^c}{\partial q^a} + \frac{\partial p^a}{\partial Q^e} \frac{\partial Q^c}{\partial p^a} = \frac{\partial Q^c}{\partial Q^a} = \delta_c^a$$

Symplectic group

We saw that symplectic form

$$\omega = \sum_a dq^a \wedge dp^a = \sum_{ab} J_{ab} d\eta^a \wedge d\eta^b$$

plays crucial role in formulation of the hamiltonian formalism.

— ω must be invariant under canonical transformation

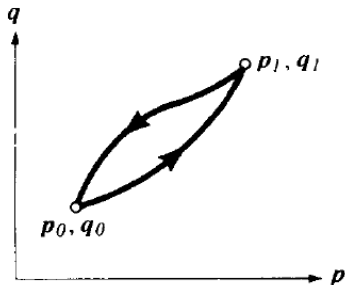
****Mathematics:** All transformations which have this property form a group
 $\mathrm{Sp}(2n, \mathbb{R})$

$$\forall M \in \mathrm{Sp}(2n, \mathbb{R}), \quad M^T J M = J, \quad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$$

$$\forall \xi, \eta, \quad [M\xi, M\eta] = [\xi, \eta]$$

Example 4

Assume that we have a closed contour ∂S in the phase space formed by two trajectories in the phase space:



Demonstrate that the integral $\oint \mathbf{p} d\mathbf{q}$ remains invariant under time-independent canonical transformation

$$\oint_{\partial S} \mathbf{p} d\mathbf{q} = \oint_{\partial S^*} \mathbf{P} d\mathbf{Q}$$

Example 4 (proof)

Note that $\mathbf{d}(\sum_a p^a dq^a) = -\sum_a dq_a \wedge dp_a = -\omega$ (symplectic form), and also due to Stokes' theorem

$$\oint_{\partial S} \mathbf{p} d\mathbf{q} = \int_S \mathbf{d} \left(\sum_a p^a dq^a \right) = - \int_S \omega$$

Since ω is invariant under canonical transformations, $\omega = \sum_a dq_a \wedge dp_a = \sum_a dQ_a \wedge dP_a$, we may apply again Stoke's theorem and get

$$\oint_{\partial S} \mathbf{p} d\mathbf{q} = - \sum_a \int_{S^*} dQ_a \wedge dP_a = \oint_{\partial S^*} \mathbf{P} d\mathbf{Q}$$

$$\mathbf{d}(\mathbf{p} d\mathbf{Q} - \mathbf{P} d\mathbf{Q}) = 0$$

so using Poincare's theorem conclude that it is exact,

$$\mathbf{p} d\mathbf{Q} - \mathbf{P} d\mathbf{Q} = dF$$

Generating function

- We saw how to *check* rapidly that transformation is canonical
- Now we'll see how to *construct* the canonical transformations which automatically satisfies all the requirements
- Minimal action principle should be valid in any set of coordinates, (p, q) or (P, Q)

$$\begin{aligned}\delta S &= \delta \int dt \left(\sum_a p_a \dot{q}_a - H(p, q, t) \right) = \\ &= \delta \int dt \left(\sum_a P_a \dot{Q}_a - \tilde{H}(P, Q, t) \right) = \min\end{aligned}$$

However in general we can't assume that integrands are equal

—they might differ by factor dF/dt

*as You remember adding total derivative to lagrangian doe not affect equations of motion

$$\begin{aligned}\Rightarrow \left(\sum_a p_a \dot{q}_a - H(p, q, t) \right) &= \\ \left(\sum_a P_a \dot{Q}_a - \tilde{H}(P, Q, t) \right) &+ \frac{d}{dt} F(q, Q, t)\end{aligned}$$

Generating function

$$\Rightarrow \left(\sum_a p_a \dot{q}_a - H(p, q, t) \right) = \left(\sum_a P_a \dot{Q}_a - \tilde{H}(P, Q, t) \right) + \frac{d}{dt} F(q, Q, t)$$

where $F(q, Q, t)$ is some function

– May rewrite this as:

$$dF = \sum p_a dq_a - \sum P_a dQ_a + (\tilde{H} - H) dt$$

– Conclude that

$$\Rightarrow p_a = \frac{\partial F}{\partial q_a}, \quad P_a = -\frac{\partial F}{\partial Q_a}, \quad \tilde{H} = H + \frac{\partial F}{\partial t}$$

– Function $F(q, Q)$ is called generating function, it allows to relate (implicitly) old and new coordinates

Example

Assume that the generating function $F_1(q, Q, t) = \sum_i q_i Q_i$. Write out explicitly the canonical transformation

Solution:

$$p_i = \frac{\partial F_1}{\partial q_i} = Q_i, \quad P_i = -\frac{\partial F_1}{\partial Q_i} = -q_i$$

so can rewrite this as

$$Q_i = p_i, \quad P_i = -q_i$$

Generating function

$$dF = \sum p_a dq_a - \sum P_a dQ_a + \tilde{H} dt - H dt$$

- Note that can rewrite

$$p_a dq_a = d(p_a q_a) - q_a dp_a$$

Let's introduce now

$$d\tilde{F} = \sum q_a dp_a - \sum P_a dQ_a + \tilde{H} dt - H dt$$

$$\tilde{F} = \tilde{F}(p, Q, t) = F - \sum_a p_a q_a$$

$\Rightarrow$ We could use $\tilde{F}$ as another generating function which depends on variables p, Q, t

$$P_a = -\frac{\partial \tilde{F}}{\partial Q_a}, \quad q_a = \frac{\partial \tilde{F}}{\partial p_a}, \quad \tilde{H} = H + \frac{\partial \tilde{F}}{\partial t}$$

Example

Write out the canonical transformations (1) for the case when generating functional F depends on other possible combinations of old and new variables: $F_2(q, P, t)$, $F_3(p, Q, t)$, $F_4(p, P, t)$.

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a}, \quad Q_a = \frac{\partial F_2}{\partial P_a} \quad (2)$$

$$\Rightarrow q_a = -\frac{\partial F_3}{\partial p_a}, \quad P_a = -\frac{\partial F_3}{\partial Q_a} \quad (3)$$

$$\Rightarrow q_a = -\frac{\partial F_4}{\partial p_a}, \quad Q_a = \frac{\partial F_4}{\partial P_a} \quad (4)$$

Example - 1

Find the generating function of the *identical* canonical transformation

$$q_a = Q_a, \quad p_a = P_a$$

Assume that corresponding generating function is $F_2(q, P, t)$

►Corresponding canonical transformations:

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a} = P_a, \quad Q_a = \frac{\partial F_2}{\partial P_a} = q_a$$

Check that equations are compatible:

$$\frac{\partial^2 F_2}{\partial q_a \partial P_b} = \frac{\partial^2 F_2}{\partial P_b \partial q_a} = \delta_{ab}$$

$\Rightarrow$ transformation is canonical (as expected), so can solve them together:

$$\Rightarrow F_2 = q_a P_a + \text{const}$$

Example 2

Find the values of α, β for which the transformation

$$q_a = \alpha Q_a, \quad p_a = \beta P_a$$

is canonical and find the generating function for this case

►Corresponding canonical transformations:

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a} = \beta P_a, \quad Q_a = \frac{\partial F_2}{\partial P_a} = \frac{1}{\alpha} q_a$$

Request that equations are compatible:

$$\frac{\partial^2 F_2}{\partial q_a \partial P_a} = \frac{\partial^2 F_2}{\partial P_a \partial q_a} \Rightarrow \beta = \frac{1}{\alpha}$$

$\Rightarrow$ can solve them together for $\beta = 1/\alpha$:

$$\Rightarrow F_2 = \frac{1}{\alpha} q_a P_a + \text{const}$$

Generating function

Find the generating function of the canonical transformation

$$Q = \log \left(\frac{\sin p}{q} \right), \quad P = q \cot p$$

Solution:

We already checked that transformation is Canonical, so focus on construction of generating function

–From definition of canonical transformation:

$$p = \arcsin(q e^Q) = \frac{\partial F(q, Q, t)}{\partial q}, \quad P = \sqrt{e^{-2Q} - q^2} = -\frac{\partial F(q, Q, t)}{\partial Q}$$

Can check that these equations are compatible, evaluation of the second derivative $\partial^2 F / \partial q \partial Q$ from both gives the same result:

$$\frac{\partial^2 F(q, Q, t)}{\partial q \partial Q} = \frac{q e^Q}{\sqrt{1 - q^2 e^{2Q}}} \Rightarrow \text{system is compatible}$$

$$F = \int dq \frac{\partial F(q, Q)}{\partial q} = q \arcsin(q e^Q) + \sqrt{e^{-2Q} - q^2}$$

Example: Harmonic oscillator

Assume that the generating function is given by

$$F(q, Q) = \frac{m\omega q^2}{2} \cot Q$$

Find explicitly the canonic transformation which it generates and the hamiltonian of harmonic oscillator $H(P, Q)$ after this transformation

Caution: We'll see that transformation is NOT trivial and allows to drastically simplify analysis/solution of equations of motion

Example: Harmonic oscillator

$$\text{Hamiltonian : } H = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}$$

$$\text{Transformation : } p = \frac{\partial F}{\partial q} = m\omega q \cot Q, \quad P = -\frac{\partial F}{\partial Q} = \frac{m\omega q^2}{2 \sin^2 Q}$$

$$\text{"solve" algebraically to get } (q, p) : \quad q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2m\omega P} \cos Q$$

Hamiltonian after transformation:

$$H_{\text{new}}(P, Q) = H_{\text{old}}(p(P, Q), q(P, Q)) + \underbrace{\frac{\partial F}{\partial t}}_{=0} = \omega P$$

-In new variables there is no dependence on Q at all, i.e. Q is cyclic variable !!!

$$\text{equations of motion : } \dot{P} = -\frac{\partial H_{\text{new}}}{\partial Q} = 0, \quad P = \text{const} = E/\omega$$

$$\dot{Q} = \frac{\partial H_{\text{new}}}{\partial P} = \omega \quad \Rightarrow \quad Q = \omega t + \text{const}$$

Example: Harmonic oscillator

Transformation : $q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2m\omega P} \cos Q$

Trajectory : $P = \text{const} = E/\omega, \quad Q = \omega t + \text{const} \quad (1)$

Compare with result (p, q) variables:

$$q = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \phi_0), \quad p = \sqrt{2mE} \cos(\omega t + \phi_0) \Rightarrow \text{agrees with } (1)$$

– Use of canonical transformation drastically simplified solution: instead of coupled ODEs

$$\dot{p} = -m\omega^2 q, \quad \dot{q} = p/m \quad [\ddot{q} + \omega^2 q = 0]$$

solved MUCH simpler decoupled system (1a,1b).

* Such variables (P, Q) are called “action-angle” variables; please see Goldstein, Landau for more details.

** Exist for any system (even multidimensional)

** Drastically simplify analysis: always Q_a are cyclic:

$$H = H(P_1, \dots, P_N)$$

$$\dot{Q}_a = \text{const} = \alpha_a \Rightarrow Q_a = \alpha_a t + \text{const}$$

Action-angle variables (formal definitions)

Consider conservative system moving in potential field $U(q)$. Assume that the trajectory is closed, and define:

$$S_0(q, E) = \int^q p dq, \quad I(E) = \oint \frac{p dq}{2\pi}$$

- S_0 is the abbreviated action defined earlier
- p is momentum. For conservative system $E = \text{const}$, so

$$p = p(q, E) = \sqrt{2m(E - U(q))}$$

- Let's consider now $S_0(q, E) = S_0(q, I)$ as a generating function of a canonical transformation $F_2(q, P)$, in which I plays the role of new canonical momentum P :

$$p = \frac{\partial F_2}{\partial q} = \frac{\partial S_0}{\partial q}, \quad Q \equiv w = \frac{\partial F_2}{\partial I} = \frac{\partial S_0}{\partial I}$$

- variable I is called action, w is angle

Action-angle variables (formal definitions)

Consider conservative system, assume that the trajectory is closed, and define:

$$S_0(q, E) = \int^q p dq, \quad I(E) = \oint \frac{p dq}{2\pi}, \quad Q \equiv w = \frac{\partial S_0}{\partial I}$$

- variable I is called action, w is angle
- Since Hamiltonian (energy E) depends only on canonical momentum but not on coordinate w , the variable w is cyclical, and

$$\dot{I} = -\frac{\partial H}{\partial w} = 0, \quad \dot{w} = \frac{dH}{dI} = \frac{dE}{dI} = \text{const}$$

$$\Rightarrow w = \frac{dE}{dI} t + \text{const}$$

We'll show now that the canonical transformation which we applied earlier to Harmonic oscillator is just transformation to action-angle variables.

Harmonic oscillator in action-angle variables

abbreviated action(gener. function) : $S_0 = \int \sqrt{2m(\omega P - kq^2/2)} dq$

$$p = \frac{\partial S_0}{\partial q} = \sqrt{2m \left(\omega P - \frac{k q^2}{2} \right)}, \quad Q = \frac{\partial S_0}{\partial P} = \omega \frac{\partial S_0}{\partial E} = \arctan \left(\frac{\sqrt{k q^2}}{\sqrt{2\omega P - k q^2}} \right),$$

$$\Rightarrow \tan Q = \frac{\sqrt{k q^2}}{\sqrt{2\omega P - k q^2}} = \frac{q}{p} \sqrt{2km}, \quad q = \frac{p}{\sqrt{km}} \tan Q = \frac{p}{m\omega} \tan Q$$

$$\frac{p^2}{2m} = \left(\omega P - \frac{k q^2}{2} \right) = \left(\omega P - \frac{p^2}{2m} \tan^2 Q \right)$$

$$\frac{p^2}{2m} (1 + \tan^2 Q) = \frac{p^2}{2m \cos^2 Q} = \omega P$$

$$\Rightarrow q = \frac{p}{m\omega} \tan Q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2m\omega P} \cos Q,$$

in agreement with our previous findings

Summary

Today we:

- defined canonical transformations and discussed what conditions they should satisfy

- found that it is possible to generate the canonical transformations using generating functions

- saw a simple example (*action-angle variables for harmonic oscillator*) where canonical transformations can drastically simplify hamiltonian and dynamics

In the next lectures we'll see more examples where canonical transformations simplify analysis